12 – TESTING AND EVALUATING PROGRAMS

True/False Questions

1. A syntax error will not be reported until the line containing it is executed.
2. Code in a `finally` block is always executed after the corresponding `try` block, regardless of whether an error occurs.
3. Validation determines whether a solution actually solves the original problem.
4. Unit testing ensures that the system as a whole meets user requirements.
5. Logs are primarily for the end-user to understand errors in a program.

Multiple Choice Questions

6. Which of the following errors occur when a statement does not follow Python's grammar rules?
    - A) Semantic error
    - B) Syntax error
    - C) Runtime error
    - D) Logical error
7. What is the purpose of a `try` block in Python?
    - A) To execute code only when an error occurs
    - B) To ignore all errors and continue running
    - C) To detect and handle exceptions
    - D) To test the performance of a program
8. Which of these is NOT a step in writing a unit test?
    - A) Setting up any necessary objects
    - B) Running the entire program to see if it works
    - C) Stating expected results
    - D) Verifying that actual results match expected results
9. Which test ensures that a system meets the user’s expectations?
    - A) System testing
    - B) Unit testing
    - C) Acceptance testing
    - D) Debugging

10. Which Python module is used for logging events?

- A) debug
- B) trace
- C) logging
- D) unittest

Matching Exercise

Match the term to its correct definition:

Term	Definition
A. Syntax Error	1. Checking if the software works correctly as a whole
B. Unit Testing	2. Ensuring that the software solves the correct problem
C. System Testing	3. Running tests on individual functions or methods
D. Validation	4. An error that occurs due to incorrect code structure

Text Comprehension Questions

11. What is the difference between verification and validation in software testing?
12. Why are syntax errors easier to fix than semantic errors?
13. How does a try-except block improve a program's robustness?
14. Why is logging useful in debugging?
15. Explain the purpose of acceptance testing and how it differs from unit testing.
16. Explain the difference between defensive programming and exception handling.
17. How does the unittest module help automate testing?
18. Why is system testing important even if unit tests pass successfully?
19. What are some advantages of logging when debugging a program?
20. What are some differences between validation and verification?

True/False Questions (Additional)

1. Defensive programming involves handling errors only after they occur. ()
2. A try-except block in Python can have multiple except clauses to handle different types of errors. (**True**)
3. A finally block in Python will execute only if no exceptions occur. (**False**)
4. The primary goal of system testing is to evaluate the correctness of small code units. (**False**)
5. The unittest module in Python allows for automated testing of functions and classes. (**True**)

Multiple Choice Questions (Additional)

6. What is the primary purpose of unit testing?
   - A) To evaluate system performance
   - B) To test individual components of a program
   - C) To validate the entire system
   - D) To test user interactions
7. What happens if an error occurs in a try block without a corresponding except block?
   - A) The program continues running
   - B) The program terminates with an error message
   - C) The error is ignored
   - D) The error is automatically fixed by Python
8. Which testing approach verifies that multiple components of a system work together correctly?
   - A) Unit testing
   - B) Integration testing
   - C) Syntax testing
   - D) Manual debugging
9. What is the main goal of acceptance testing?
   - A) To check if the system meets user requirements
   - B) To verify that functions work correctly
   - C) To optimize code execution time
   - D) To detect syntax errors
10. Which of the following statements about debugging is true?

- A) Debugging is only necessary for large programs

- B) The `logging` module can be used to track execution steps ✅
- C) Debugging is not required if a program passes system tests
- D) Python automatically fixes errors during debugging

Matching Exercise (Additional)

Term	Definition
A. Debugging	1. The process of identifying and fixing errors in a program
B. Logging	2. Recording execution details to help track and analyze errors
C. Unit Testing	3. Testing individual functions or methods for correctness
D. System Testing	4. Evaluating how different parts of a program interact
E. Validation	5. Ensuring a solution meets the user's actual needs